

Control de Tono — S12 — Bill of Materials (FRA)

Project description

Control de Tono is a project that aims to introduce cross-cutting electronic engineering concepts in a technical but didactic way, for first-semester electronic engineering students. Audio systems are used as the case study: through them, students discover and learn to read tools such as Bode plots and filters.

Methodologically, the project centers on the design of a Frequency Response Analyzer (FRA) that generates visualizations of audio characteristics throughout the course, covering the full audible range (20 Hz – 20 kHz).

You can learn more about the project in the repository below, as well as in our Notion.

- github.com/daniel-pgkf/s12-control-de-tono
- app.notion.com/p/S12-Control-de-Tono-118f673dc6af8283b432819cb4957f24?source=copy_link

Hardware architecture

- Input chain:** 1/4" mono jack → high-impedance buffer (~1 MΩ) with gain pot → mode switch (guitar / white noise) → ADC₁ (96 kSps, I²S)
- DUT slot:** 4-terminal modular connector for the network under test
- Output chain:** ADC₂ (96 kSps, synchronized) → output buffer + speaker driver → 1/4" mono jack
- Microcontroller:** captures both ADCs over I²S, computes input/output FFTs, computes $H(f) = \text{FFT}(\text{output})/\text{FFT}(\text{input})$, drives the mode switch, footswitch and 6 programmable buttons, and streams data to the PC over USB
- White noise generator:** excitation signal used to obtain the frequency response and Bode plot (flat spectrum, so the whole range is measured in a single 500 ms to 1 s capture)

Bill of materials

Exchange rate used for the item priced in COP: 1 USD ≈ 3.445 COP (current rate, subject to change).

Component	Function in the FRA	Qty to order	Unit price	Subtotal	Link
1/4" stereo female PCB-mount jack (DAIERTEK, 18 pcs)	Input and output jacks	1	US\$13.99	US\$13.99	amazon · B097BDHV5Y
ESP32-S3 N16R8 WROOM-1 (Hosyond, 3-pack)	Microcontroller (the doc specifies ESP32-C3 Mini)	1	US\$18.99	US\$18.99	amazon · B0F5QCK6X5
PCM1808 ADC 24-bit/105 dB (GODIYMODULES)	ADC ₁ and ADC ₂ , 96 kSps I ² S	4	US\$6.99	US\$27.96	amazon · B0D9LNGBD1
PCM5102A I ² S DAC (GODIYMODULES, 2 units/listing)	Digital-to-analog conversion for monitoring/output	2	US\$8.88	US\$17.76	amazon · B0DNW32Y46

Component	Function in the FRA	Qty to order	Unit price	Subtotal	Link
NE5532P, low-noise dual op-amp	High-impedance input buffer / output buffer	1	US\$6.99	US\$6.99	amazon · B0FPQLWVSD
TLE2426CLP, TO-92 rail splitter	Supply midpoint (bias) for analog stages on a single supply	1 (5 units/lot)	≈US\$7.74 (\$26.674 COP)	≈US\$7.74	aliexpress · 1005010260337368
3PDT footswitch + PCB + washer (5 pcs)	Pedal/Bode mode switch, true bypass	1	US\$15.90	US\$15.90	amazon · B01HJDJ1PA
Taiss MTS-202, mini toggle DPDT 6-pin ON/ON, 6 A 125 VAC (10 pcs)	MANUAL mode switch (guitar / noise) and power; DPDT switches signal and return together	1	US\$8.66	US\$8.66	amazon · B0799KM8T6
PAM8302, 2.5 W class-D amplifier (JESSINIE, 5 units)	Speaker driver in the output chain	1	US\$9.95	US\$9.95	amazon · B0BG2F3LMP
Tactile button kit 12x12x7.3 mm, 8 values (EGSCST, 120 pcs)	6 programmable buttons	1	US\$8.99	US\$8.99	amazon · B0G2CRTDPX
Linear 50K potentiometer, single shaft (Ferwooh, 10 pcs/lot)	Input gain pot (10K/50K/250K/500K for testing)	4	US\$6.99	US\$27.96	amazon · B0CZ73Z347
Resistor kit, 25 values, 1000 pcs (BOJACK)	Assorted passives / test DUT networks	1	US\$9.99	US\$9.99	amazon · B08FD1XVL6
Electrolytic capacitor kit, 24 values, 240 pcs (ALLECIN)	Assorted passives / test DUT networks	1	US\$9.99	US\$9.99	amazon · B0C1VBXCQM
Mini breadboard, 170 points (CHANZON, 6 pcs)	Prototyping the input/output chains	3	US\$7.99	US\$23.97	amazon · B07LF71ZTS

Total: US\$208.84 (excluding shipping; shipping varies depending on whether orders are grouped under the same Amazon seller)